

Phrase 13

Table 1. Xception Dual-Stream Evaluation Results

Metric	Value
Accuracy	0.9637
Precision	0.9585
Recall	0.9695
F1-Score	0.9640
Specificity	0.9580
AUC-ROC	0.9944

Table 2. Confusion Matrix

	Predicted FAKE	Predicted REAL
Actual FAKE	9,695 (TP)	305 (FN)
Actual REAL	420 (FP)	9,580 (TN)

Observation	Interpretation
Accuracy	The model correctly classifies 96.37% of all test images.
Precision	A precision of 95.85% indicates that nearly all images predicted as fake are correctly classified.
Recall	A recall of 96.95% demonstrates excellent detection of fake images with very few missed samples.
F1-Score	The F1-score of 96.40% reflects an outstanding balance between precision and recall.
Specificity	The model correctly identifies 95.80% of real images, resulting in very few false positives.
AUC-ROC	An AUC-ROC of 0.9944 indicates near-perfect discrimination between real and fake images.

Key Finding	The Xception-based dual-stream architecture achieves the highest overall performance, demonstrating that combining spatial and frequency-domain features with a powerful feature extractor significantly improves deepfake detection accuracy and robustness.
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